Aleksas Mazeliauskas

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Professional experience

01/10/2019 –30/09/2021 Postdoctoral fellow at **CERN Theoretical Physics Department**. Research in heavy-ion phenomenology, thermalization and non-equilibrium QCD phenomena in small systems.

01/04/2017- -30/09/2019 Postdoctoral researcher at the **Institute for Theoretical Physics, Heidelberg University**, member of collaborative research center "Isolated quantum systems and universality in extreme conditions". Contributions to projects:

- A01: Initial state and thermalisation dynamics in heavy-ion collisions. Prof. Jürgen Berges, Prof. Klaus Reygers, and Prof. Johanna Stachel
- C05: Flow and fluctuations in relativistic heavy ion collisions. Priv.-Doz. Dr. Stefan Flörchinger and Priv.-Doz. Dr. Silvia Masciocchi

Work on universal early times dynamics of heavy ion collisions using QCD kinetic theory (pre-scaling, entropy production, chemical equilibration). Study of propagation of fluctuations in pre-equilibrium (with kinetic propagator `KøMPøST`), mode-by-mode hydrodynamics (`FluiduM`) and hadronic stages (resonance decay code `FastReso`).

25/06/2014- -31/03/2017 Research assistant at **Nucler Theory Group, Stony Brook University**, PhD dissertation work under supervision of Prof. Derek Teaney, *Fluctuations in ultra-relativistic heavy ion collisions*

- Early stages of heavy ion collisions with effective kinetic theory equilibration
- Nonlinear thermal noise corrections to heavy ion expansion
- Subleading harmonic flows and factorization breaking with principal components
- 3D viscous relativistic hydrodynamic code for heavy ion collisions `RNAVIER`

03/07/2013- -08/01/2014 Research assistant at **Nucler Theory Group, Stony Brook University**, Analyzed temperature dependent corrections to anomalous transport coefficients. With Prof. Dmitri Kharzeev and Dr. Tigran Kalaydzhyan.

Education

27/08/2012- -31/03/2017 **Stony Brook University**, United States, Doctor of Philosophy in Physics, Max Dresden Prize for outstanding theoretical thesis, degree awarded on 19/05/2017

01/10/2011- -25/06/2012 **University of Cambridge**, United Kingdom, Master of Mathematics, Part III Maths, graduated with distinction, degree awarded on 29/06/2012

01/10/2008- -25/06/2011 **University of Cambridge**, United Kingdom, Bachelor of Arts, Mathematical Tripos, 1st class degree, degree awarded on 29/06/2012

Awards

09/11/2019 Nuclear Physics A Young Scientist Award for the best theory talk at Quark Matter 2019

02/05/2017 Max Dresden Prize for outstanding theoretical thesis

25/07/2016 APS FGSA Travel Award for Excellence in Graduate Research

23/05/2013 David Fox award for the outstanding Teaching Assistant

2009-2011 Master's Sizarship for outstanding achievement by undergraduate member of College, T W Armour Prize for outstanding performance in University examination in Mathematics, John Cartwright (1674) scholarship, Drury-Johns Mathematical Prize,

2006-2007 Honourable mention in International Physics Olympiad, Iran, Vilnius International Rotary Club scholarship, Honourable mention in International Physics Olympiad, Singapore

Publications

2019

1. O. Garcia-Montero, N. Löher, A. Mazeliauskas, J. Berges, and K. Reygers, *Probing the evolution of heavy-ion collisions using direct photon interferometry*, arXiv:1909:12246
2. D. Devetak, A. Dubla, S. Floerchinger, E. Grossi, A. Mazeliauskas, S. Masciocchi, and I. Selyuzhenkov, *Global fluid fits to identified particle transverse momentum spectra from heavy-ion collisions at the Large Hadron Collider*, arXiv:1909:10485
3. P. Hanus, A. Mazeliauskas, and K. Reygers, *Entropy production in pp and Pb-Pb collisions at energies available at the CERN Large Hadron Collider*, **Phys. Rev. C 100, 064903 (2019**), arXiv:1908.02792
4. G. Giacalone, A. Mazeliauskas and S. Schlichting, *Hydrodynamic Attractors, Initial State Energy, and Particle Production in Relativistic Nuclear Collisions*, **Phys. Rev. Lett. 123, 262301 (2019)**, arXiv:1908.02866
5. A. Mazeliauskas and V. Vislavicius, *Temperature and fluid velocity on the freeze-out surface from π, K, p spectra in pp, p–Pb and Pb–Pb collisions*, arXiv:1907.11059
6. W. Florkowski, A. Kumar, R. Ryblewski, and A. Mazeliauskas, *Longitudinal spin polarization in a thermal model*, **Phys. Rev. C 100, 054907 (2019)**, arXiv:1904.00002

2018

7. A. Kurkela, A. Mazeliauskas, *Chemical equilibration in weakly coupled QCD*, **Phys. Rev. D 99, 054018 (2019)**, EDITORS' SUGGESTION, arXiv:1811.03068
8. A. Kurkela, A. Mazeliauskas, *Chemical equilibration in hadronic collisions*, **Phys. Rev. Lett. 122, 142301 (2019)**, arXiv:1811.03040
9. A. Mazeliauskas, J. Berges, *Prescaling and far-from-equilibrium hydrodynamics in the quark-gluon plasma*, **Phys. Rev. Lett. 122, 122301 (2019)**, arXiv:1810.10554
10. A. Mazeliauskas, S. Floerchinger, E. Grossi, D. Teaney, *Fast resonance decays in nuclear collisions*, **Eur. Phys. J. C (2019)**, arXiv:1809.11049
11. A. Kurkela, A. Mazeliauskas, J.-F. Paquet, S. Schlichting and D. Teaney, *Matching the non-equilibrium initial stage of heavy ion collisions to hydrodynamics with QCD kinetic theory*, **Phys. Rev. Lett. 122, 122302 (2019)**, EDITORS' SUGGESTION, arXiv:1805.01604
12. A. Kurkela, A. Mazeliauskas, J.-F. Paquet, S. Schlichting and D. Teaney, *Effective kinetic description of event-by-event pre-equilibrium dynamics in high-energy heavy-ion collisions*, **Phys. Rev. C,99, 034910 (2019)**, arXiv:1805.00961

pre-2018

13. Y. Akamatsu, A. Mazeliauskas and D. Teaney, *Bulk viscosity from hydrodynamic fluctuations with relativistic hydro-kinetic theory*, **Phys. Rev. C 97, 024902 (2018)**, arXiv:1708.05657
14. Y. Akamatsu, A. Mazeliauskas and D. Teaney, *A kinetic regime of hydrodynamic fluctuations and long time tails for a Bjorken expansion*, **Phys. Rev. C,95, 014909 (2017)**, arXiv:1606.07742

15. L. Keegan, A. Kurkela, A. Mazeliauskas and D. Teaney, *Initial condition for hydrodynamics from weakly coupled pre-equilibrium evolution*, **JHEP 1608 (2016) 171**, arXiv:1605.04287
16. A. Mazeliauskas and D. Teaney, *Fluctuations of harmonic and radial flow in heavy ion collisions with principal components*, **Phys. Rev. C 93, 024913 (2016)**, arXiv:1509.07492
17. A. Mazeliauskas and D. Teaney, *Subleading harmonic flows in hydrodynamic simulations of heavy ion collisions*, **Phys. Rev. C 91, 044902 (2015)**, arXiv:1501.03138

Organized workshops

19/08/2019–23/08/2019 Convener of discussion group on "Connections between QGP-like observables in small systems" at **International Workshop on QCD Challenges from pp to AA**, Lund, Sweden

08/04/2019–12/04/2019 Convener of discussion group on "Implementations of critical dynamics" at **EMMI Rapid Reaction Task Force: Dynamics of critical fluctuations**, Darmstadt, Germany

Invited presentations

06/12/2019 talk at **XIX Zimányi School Winter Workshop** on *Pre-equilibrium phenomena in Quark Gluon Plasma*, Budapest, Hungary

25/11/2019 talk at **Theoretical Foundations of Relativistic Hydrodynamics Workshop** on *Pre-scaling, hydrodynamic attractors and entropy production in heavy ion collisions*, Banff, Canada

01/11/2019 talk at **New Development of Hydrodynamics and its applications in Heavy-ion Collisions Workshop** on *Hydrodynamic attractors, initial state energy and particle production in relativistic nuclear collisions*, Shanghai, China

13/09/2019 plenary talk at **XLIX International Symposium on Multiparticle Dynamics** on *Equilibration in Quark Gluon Plasma*, Santa Fe, New Mexico, USA

25/06/2019 plenary talk at **V Initial Stages Conference** on *Matching of initial conditions to hydro evolution*, New York, USA

06/04/2019 talk at **XIV Polish Workshop on Relativistic Heavy-Ion Collisions** on *Chemical equilibration in hadronic collisions*, Krakow, Poland

01/08/2018 parallel talk at **XIII Quark Confinement and the Hadron Spectrum Conference**, *2+1D simulations of pre-equilibrium stage with QCD kinetic theory*, Dublin, Ireland

18/05/2018 plenary talk at **XXVII Quark Matter Conference** on *Initial conditions for nuclear collisions: theory overview*, Venice, Italy

02/10/2017 talk at **XII High-pT Physics Workshop** on *Initial conditions for heavy ion collisions* , Bergen, Norway

15/06/2017 talk at **XII Workshop on Particle Correlations and Femtoscopy** on *Kinetic theory equilibration for realistic heavy ion initial conditions*, Amsterdam, The Netherlands

06/03/2015 talk at **Collectivity in Small Colliding Systems with High Multiplicity Workshop** on *Principal Component Analysis and Subleading Flow*, Brookhaven National Lab, New York, United States

Contributed presentations

06/11/2019 **XXVIII Quark Matter Conference**, Wuhan, China

22/09/2019 **Quantum Systems in Extreme Conditions**, Heidelberg, Germany

10/06/2019 **XVIII Strangness in Quark Matter**, Bari, Italy

14/01/2019 **XLVII Workshop on Gross Properties of Nuclei and Nuclear Excitations**, Hirschegg, Austria

23/05/2019 **Origins of Correlations in High Energy Collisions**, INT, Seattle, Washington, United States

09/04/2019 **EMMI RRTF: Dynamics of critical fluctuations**, Darmstadt, Germany

02/10/2018 **IX Hard Probes conference**, Aix-Les-Bains, France

27/08/2018 **Probing the Quark-Gluon Plasma with collective phenomena and heavy quarks**, MIAPP, Munich, Germany

19/09/2017 **IV Initial Stages**, Krakow, Poland

08/08/2017 **Critical Point and Onset of Deconfinement**, Stony Brook, New York, United States

07/02/2017 **XXVI Quark Matter**, Chicago, Illinois, United States

19/09/2016 **VII Hot Quarks**, South Padre Island, Texas, United States

25/05/2016 **III Initial Stages**, Lisbon, Portugal

22/07/2015 **Correlations and Fluctuations in p+A and A+A Collisions**, INT, Seattle, Washington, United States

27/03/2015 **Ohio-Region APS meeting**, Kent, Ohio, United States

Teaching and mentoring experience

2019 Informal mentoring of PhD students Oscar Garcia-Montero, Giuliano Giacalone and Aleksandr Mikheev.

2018 **Co-supervisor** of a bachelor student (Paul Wiesemeyer) project on radiative photon production in QGP (summer semester).

2018 **Co-organiser** of a master level student lead seminar on QCD matter in heavy ion collisions (summer semester).

2017-2019 **Co-head tutor** for Quantum Field Theory course. Responsible for preparing homework and exam problems and overseeing other tutors (winter and summer semesters).

2014 **Recitation instructor** for calculus based undergraduate electromagnetism and mechanics courses (2 semesters). Selected student responses: *I think the instructor was more valuable than the course itself.*; *I would love to see him become a professor one day and attain his career goals.*; *He would do a fantastic job in teaching the entire course.*

2012–2013 **Teaching assistant** for undergraduate mechanics laboratory (2 semesters), named outstanding Teaching Assistant.

Science outreach

2019 Interviewed about my work for French science magazine: J.-B. Veyrieras, *Découverte de la première flèche du temps*, Science & Vie, n°1228 (2020)

Prepared physics questions for an on-line science quiz competition *Išmanioji diena* held on 10/10/2019, Lithuania

2017 Prepared two introductory problem sets on elementary particles for high school students at Vilnius University Particle physics outreach group

2008-2013 One of founders and organisers of student-run physics team competition at Vilnius Lyceum, created a number of physics problems for the competition.

2007-2012 Active alumnus of NMA – additional training school for gifted youth in Lithuania; spoke twice on Lithuanian national TV about education of talented children.

Languages

Lithuanian	native, enthusiastic teacher	**German**	upper intermediate
English	fluent, higher education in English	**Russian**	basic conversational
Chinese	beginner, passed HSK level II test	**French**	beginner

Programming

Programming	Extensive numerically oriented coding in `C++` and `Python`, previous experience with `Fortran` and Matlab, good working knowledge of Mathematica
General IT	Git (source code management), LaTeX (word processing), Inkscape (vector graphics), GNU/Linux (operating system), Gnuplot (graphing), HTML5 (web building)